

POWER BI PROJECT REPORT

Bank Customer Analytics & Financial Insights

Power BI Dashboard and Analytical Question Report

Project Information	Details
Project domain	Banking / Financial Analytics
Tool	Microsoft Power BI
Primary analysis entity	All Customer Transaction
Report structure	1 Dashboard + Q.6–Q.24 analytical pages
Custom visuals	Violin Plot, Box Plot, Box and Whisker
Scope	Customer, account, transaction, branch, credit, loan and risk analysis

1. Executive Summary

This project uses Microsoft Power BI to transform banking customer and transaction data into an interactive analytical report. The uploaded PBIX contains a dedicated dashboard and analytical pages labelled Q.6 through Q.24. The analysis covers account balances, transaction amounts and types, transaction dates and times, branches, currencies, credit scores, loan amounts, interest rates, employment sectors, cities and customer risk indicators.

The central purpose is to demonstrate how business data can be explored through interactive visualizations, aggregation, filtering, segmentation and comparative analysis.

Key analytical themes

- Customer and account profiling
- Transaction volume, value and timing
- Branch-level performance comparison
- Credit-score and loan analysis
- Interest-rate and balance relationship analysis
- Risk-level and normalized-risk analysis
- City and employment-sector segmentation
- Interactive filtering using slicers

2. Project Objectives

- Build an interactive banking analytics dashboard in Power BI.
- Analyze customer accounts and financial balances.
- Measure transaction counts, amounts and transaction values.
- Compare branches, cities, currencies and account types.
- Study credit-score ranges and lending variables.
- Explore the relationship between interest rates and account balances.
- Examine transaction activity across dates, months, years and transaction times.
- Classify customers/accounts using risk-related indicators.
- Use slicers and interactive visuals for exploratory analysis.
- Present the analysis for academic and portfolio use.

3. Data Description

The PBIX report uses a primary analytical entity named “All Customer Transaction” (spelling retained from the model). The report definitions show fields spanning customer identity, accounts, transactions, branches, credit, loans, employment and risk.

Category	Fields
Customer / Account	AccountNumber, AccountHolder, AccountType, Customer(year), OpeningDate, Balance(USD)
Transaction	TransactionID, TransactionType, TransactionDate, TransactionTime(hr), Amount(USD), TransactionValue
Location / Branch	BranchCode, City
Financial	Currency, Amount, Amount(USD), Balance(USD), Loan Amount(USD), InterestRate
Credit	CreditScore, Credit Score Range, CreditScore (bins)
Employment	Employment Sector
Risk	Risk Level, NormalizedRiskScore, AverageTransactionAmount

4. Methodology

Stage	Implementation
Data preparation	The banking dataset is represented in the Power BI model and exposed through report visuals and fields.
Data modelling	The report uses the main customer-transaction entity and measures/aggregations such as SUM, COUNT and AVERAGE.
Data segmentation	Fields such as account type, transaction type, currency, branch, risk level and credit-score range are used to segment the data.
Visualization	The project uses cards, tables, charts, slicers, a funnel, ribbon charts, a treemap, a scatter chart, a donut chart and custom statistical visuals.
Interaction	Slicers and cross-filtering allow users to investigate the same dataset from different dimensions.
Interpretation	The dashboard provides a high-level view, while Q.6–Q.24 pages address specific analytical questions.

5. Dashboard Design

The main page is named “DASHBOARD” and is configured as a 1500 × 800 report canvas. It contains KPI cards, slicers and multiple analytical visuals.

Component	Purpose
KPI cards	Transaction count, Credit Score, Total Loan Amount, Total Balance and transaction amount measures.
Slicers	Account Type, Transaction Date, Branch Code and Currency.
Trend visual	Transaction activity by transaction date.
Distribution visual	Account/customer distribution using account number and customer-year information.
Funnel	Transaction counts segmented by Account Type.
Column chart	Account counts/records by Opening Date year.
Ribbon chart	Transaction counts by City and Employment Sector.
Analytical table	Credit Score Range with Interest Rate and Loan Amount(USD).

The dashboard is designed for interactive exploration. Selecting an account type, branch, currency or date can narrow the visual context and support focused investigation.

6. Question-wise Analysis

The uploaded PBIX contains analytical pages Q.6 through Q.24. The following descriptions are based on the fields and visual configurations embedded in the report definition. Numerical conclusions are not invented where the PBIX metadata does not expose rendered result values.

Page	Topic	Visuals	Analytical purpose
Q.6	Account Type & Balance	Table and clustered bar chart	Compare account types using balance-related measures.
Q.7	Currency & Amount	Clustered bar charts	Compare average transaction/amount measures across currencies.
Q.8	Branch Transactions	Clustered column chart	Compare transaction counts across BranchCode.
Q.9	Interest Rate vs Balance	Scatter chart + correlation card	Examine the relationship between InterestRate and Balance(USD), including a correlation coefficient.
Q.10	Credit Score & Loans	Donut chart	Compare loan amount across Credit Score Range categories.
Q.11	Transaction Trends	Pivot table + line/column charts	Study transaction activity across transaction dates and transaction types.
Q.12	Customer Year	Tables	Review account/customer information by Customer(year).

Page	Topic	Visuals	Analytical purpose
Q.13	Transaction Type & Value	Pivot/bar visuals	Compare transaction counts and transaction value by transaction type.
Q.14	Transaction Time	Treemap + slicer	Analyze transaction counts by TransactionTime(hr) and transaction type.
Q.15	Credit Score Distribution	Column chart	Examine customers/accounts across credit-score bins/ranges.
Q.16	Customer & Balance	Tables	Inspect account holder, account number, customer year and balance details.
Q.17	Branch Performance	Table	Compare branch transaction count, amount and a branch-rating measure.
Q.19	Risk Analysis	Column chart + table	Compare Risk Level and inspect risk-related financial indicators.
Q.20	City & Employment	Ribbon chart	Compare transaction counts across cities and employment sectors.
Q.21	Account vs Transaction Type	Bar chart + card + slicer	Compare transaction activity by account type and transaction type.
Q.22	Transaction Detail	Slicer + line chart	Use AccountNumber filtering to investigate transaction activity over time.
Q.23	City & Employment	Ribbon chart	Analyze the interaction of city and employment-sector segments.
Q.24	Account-level Financials	Table	Review account number, transaction amount and balance together.

7. Detailed Analytical Interpretation

7.1 Balance and account analysis

Account type and account-level pages allow balances to be compared across customer/account segments. This is useful for identifying how financial balances are distributed and for investigating individual accounts when a particular account number is selected.

7.2 Transaction analysis

Transaction ID counts, transaction amounts, transaction values and transaction types form a major part of the report. The project therefore supports both volume analysis (how many transactions occur) and value analysis (how much money is involved).

7.3 Time analysis

TransactionDate and TransactionTime(hr) are used to investigate when banking activity occurs. The report includes date hierarchies and time-oriented visuals, enabling comparisons by year, month and transaction time.

7.4 Branch and geographic analysis

BranchCode and City provide geographic dimensions for comparison. Branch-level transaction counts and amounts can be used to identify differences in activity between locations, while city-level visuals show broader geographic patterns.

7.5 Credit and lending analysis

CreditScore, Credit Score Range and CreditScore (bins) are used alongside Loan Amount(USD). This structure enables the analyst to compare lending exposure across credit segments without relying on a single average.

7.6 Interest-rate analysis

The Q.9 scatter chart places InterestRate against Balance(USD), while a correlation coefficient is displayed as a separate measure. This supports a quantitative examination of whether the two variables move together in the dataset.

7.7 Risk analysis

Risk Level, NormalizedRiskScore and AverageTransactionAmount are brought together on Q.19. This supports customer/account segmentation by risk category and enables financial variables to be viewed alongside risk indicators.

7.8 Employment-sector and city analysis

Ribbon charts compare transaction counts by City and Employment Sector. This provides a multidimensional view of transaction activity and helps reveal how customer employment segments differ across locations.

8. Power BI Visualizations Used

Visual	Use	Project application
Card	KPI-style totals and measures	Dashboard and analytical pages
Slicer	Interactive filtering	Account Type, Date, Branch, Currency, Transaction Type, Account Number
Table	Detailed account/transaction inspection	Multiple question pages
Clustered column/bar	Category comparison	Branch, currency, risk, credit and account analyses
Line chart	Time-series analysis	Transaction trends
Donut chart	Category distribution/value comparison	Credit-score/loan and customer distribution
Treemap	Hierarchical/category activity	Transaction time/type
Scatter chart	Relationship/correlation	Interest rate vs. balance
Ribbon chart	Rank/segment comparison	City and employment sector
Funnel	Stage/category volume comparison	Account type and transactions
Pivot table	Cross-tab analysis	Transaction date/type and credit/lending analysis
Custom statistical visuals	Distribution and spread analysis	Violin plot, box plot, box-and-whisker visuals available in the PBIX

9. Measures and Calculations

The report definition confirms model measures and aggregations including total loan amount, total balance, transaction counts and a coefficient-correlation measure. Visuals also use SUM and AVERAGE aggregations for financial amounts and count-based aggregations for activity.

Measure / field	Concept	Purpose
Total Balance	SUM(Balance(USD))	Aggregates account balances.
Total Loan Amount	SUM(Loan Amount(USD))	Aggregates loan exposure.
Transaction Count	COUNT / non-null count of TransactionID	Measures transaction activity.
Average Transaction Amount	AVERAGE(Amount(USD))	Measures typical transaction size.
Correlation Coefficient	Model measure: coeff corr	Quantifies the relationship used in Q.9.
Branch Rating	DenseBranchRating	Used in the branch-performance analysis.
Risk Score	NormalizedRiskScore	Used to support risk segmentation.

10. Key Insights Supported by the Dashboard

- The project combines customer, account and transaction perspectives.
- The dashboard supports both high-level KPIs and detailed account-level investigation.
- Branch, city, currency, account type and employment sector provide multiple segmentation dimensions.
- Credit-score ranges can be compared with loan amounts.
- Transaction dates and times support temporal activity analysis.
- Risk level and normalized risk provide a dedicated risk dimension.
- The interest-rate versus balance scatter plot and correlation measure provide relationship analysis.

- Interactive slicers make the report suitable for exploratory analysis.

11. Limitations and Considerations

- PBIX metadata identifies fields and calculations but does not expose every rendered chart value in a directly readable form; exact numerical findings should be taken from the live Power BI report.
- Correlation does not establish causation.
- Risk categories depend on the definitions and thresholds encoded in the model.
- Results can change when slicers or filters are applied.
- Data quality, missing values, duplicates and source definitions should be validated before operational use.

12. Conclusion

The Bank Customer Analytics project demonstrates the use of Microsoft Power BI to convert banking data into an interactive analytical solution. The uploaded PBIX combines a management-style dashboard with question-specific analytical pages covering customer accounts, transactions, branches, credit scores, loans, interest rates, risk, geography and employment sectors.

The project demonstrates data aggregation, segmentation, filtering, time analysis, relationship analysis, KPI reporting and detailed tabular investigation. A production-oriented version should additionally validate data quality, document business definitions and establish appropriate security and refresh procedures.

Appendix A – Page Inventory

Page	Description
DASHBOARD	Executive dashboard
Q.6	Account-type and balance analysis
Q.7	Currency and transaction amount analysis
Q.8	Branch-wise transaction count
Q.9	Interest rate vs. balance relationship
Q.10	Credit-score range and loan-amount analysis
Q.11	Transaction trends over time
Q.12	Customer/account details by customer year
Q.13	Transaction type and transaction value
Q.14	Transaction time and transaction-type analysis
Q.15	Credit-score distribution
Q.16	Customer/account details and balance
Q.17	Branch-wise transaction performance
Q.19	Risk-level distribution and risk metrics
Q.20	City and employment-sector transaction analysis
Q.21	Account type vs. transaction type
Q.22	Transaction/account detail analysis
Q.23	City and employment-sector transaction analysis
Q.24	Account-level amount and balance analysis

Appendix B – Field Inventory

Field	Field	Field
AccountNumber	AccountHolder	AccountType

Field	Field	Field
Balance(USD)	Amount(USD)	TransactionID
Field	Field	Field
TransactionType	TransactionDate	TransactionTime(hr)
Field	Field	Field
OpeningDate	Customer(year)	BranchCode
Field	Field	Field
City	Currency	CreditScore
Field	Field	Field
Credit Score Range	CreditScore (bins)	Loan Amount(USD)
Field	Field	Field
InterestRate	Employment Sector	Risk Level
Field	Field	Field
NormalizedRiskScore	AverageTransactionAmount	TransactionValue

Appendix C – Submission Note

This report was prepared from the uploaded Power BI project file. Page names, major fields, visual types and dashboard structure were taken from the PBIX report definition. Where exact rendered numerical results were not directly recoverable, the report describes analytical purpose instead of fabricating values.